

Student's Name: _____ Teacher's Code: _____



Saint Ignatius' College, Riverview
Mathematics Assessment Task
2025

Year 12
Mathematics (Extension One)
Task 4
Trial HSC Examination
Date: Wednesday 13th August 2025

General Instructions:

- **Reading time:** 10 minutes
- **Time Allowed:** 2 hours.
- Write using a black pen.
- Calculators approved by NESA may be used.
- Attempt all questions in the booklets provided.
- Write **your name** and **your teacher's code** in the positions indicated.
- Marks may not be awarded for missing or carelessly arranged work.

Teachers:

- | | |
|----------------|------------|
| • Mr R Maxwell | REM |
| • Mr D Reidy | DPR |
| • Mr S Maher | SJM |
| • Mr P Collins | PPC |

Topics Examined:

Section A **10 Marks**
Multiple Choice

Section B

Short Answer

Question 11	15 Marks
Question 12	15 Marks
Question 13	15 Marks
Question 14	15 Marks

Total **70 Marks**

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SECTION A: Multiple Choice Questions 1 - 10:**MARKS 10**

1. The multiplicity of the negative root of the polynomial $y = (x - 3)^3(x - 2)(x + 4)^4$ is

- A. 1
- B. 2
- C. 3
- D. 4

2. The magnitude of the projection of the vector $\vec{a} = \vec{i} + 4\vec{j}$ in the direction of

the vector $\vec{b}$ is $\frac{11}{\sqrt{13}}$.

Which of the following could be equal to $\vec{b}$?

- A. $2\vec{i} + 3\vec{j}$
- B. $3\vec{i} + 2\vec{j}$
- C. $3\vec{i} + 4\vec{j}$
- D. $4\vec{i} + 3\vec{j}$

3. What is the exact value of $\sin 75^\circ$

- A. $\frac{1 + \sqrt{3}}{2\sqrt{2}}$
- B. $\frac{1 - \sqrt{3}}{2\sqrt{2}}$
- C. $\frac{\sqrt{2} - \sqrt{6}}{2}$
- D. $\frac{\sqrt{2} + \sqrt{6}}{2}$

4. The value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is:

A. $\frac{\pi}{3}$

B. $-\frac{\pi}{3}$

C. $\frac{2\pi}{3}$

D. $\frac{4\pi}{3}$

5. Using the trigonometric product to sum result, the value of $\int \sin 3x \sin x dx$ is:

A. $\frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + c$

B. $\frac{1}{8} \sin 2x - \frac{1}{4} \sin 4x + c$

C. $\frac{1}{4} \cos 2x - \frac{1}{8} \cos 4x + c$

D. $\frac{1}{8} \cos 4x - \frac{1}{4} \cos 2x + c$

6. Consider the equation $x^2 + 2x + 3 = 0$. The sum of each of the reciprocal roots of this equation are:

A. $\frac{2}{3}$

B. $-\frac{2}{3}$

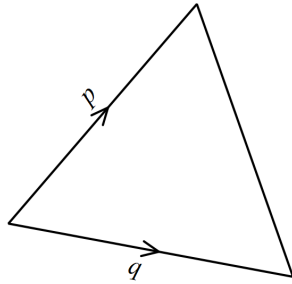
C. $\frac{1}{2}$

D. $-\frac{1}{2}$

7. If $f(x) = \frac{3 + e^{2x}}{5}$, the expression for $f^{-1}(x)$ is:

- A. $\ln(5x - 3)$
- B. $\frac{1}{2}\ln(5x - 3)$
- C. $\ln 5x - \ln 3x$
- D. $\frac{1}{2}(\ln 5x - \ln 3x)$

8. An equilateral Δ of side length 3 units is shown below.



Vectors $\vec{p}$ and $\vec{q}$ are represented in the diagram, the value of $\vec{p} \cdot \vec{q}$ is:

- A. 9
- B. $\frac{9}{\sqrt{2}}$
- C. 0
- D. $\frac{9}{2}$

9. Given that $x = 40t$ and $y = 56t - 16t^2$, then the expression for $\frac{dy}{dx}$ is given by:

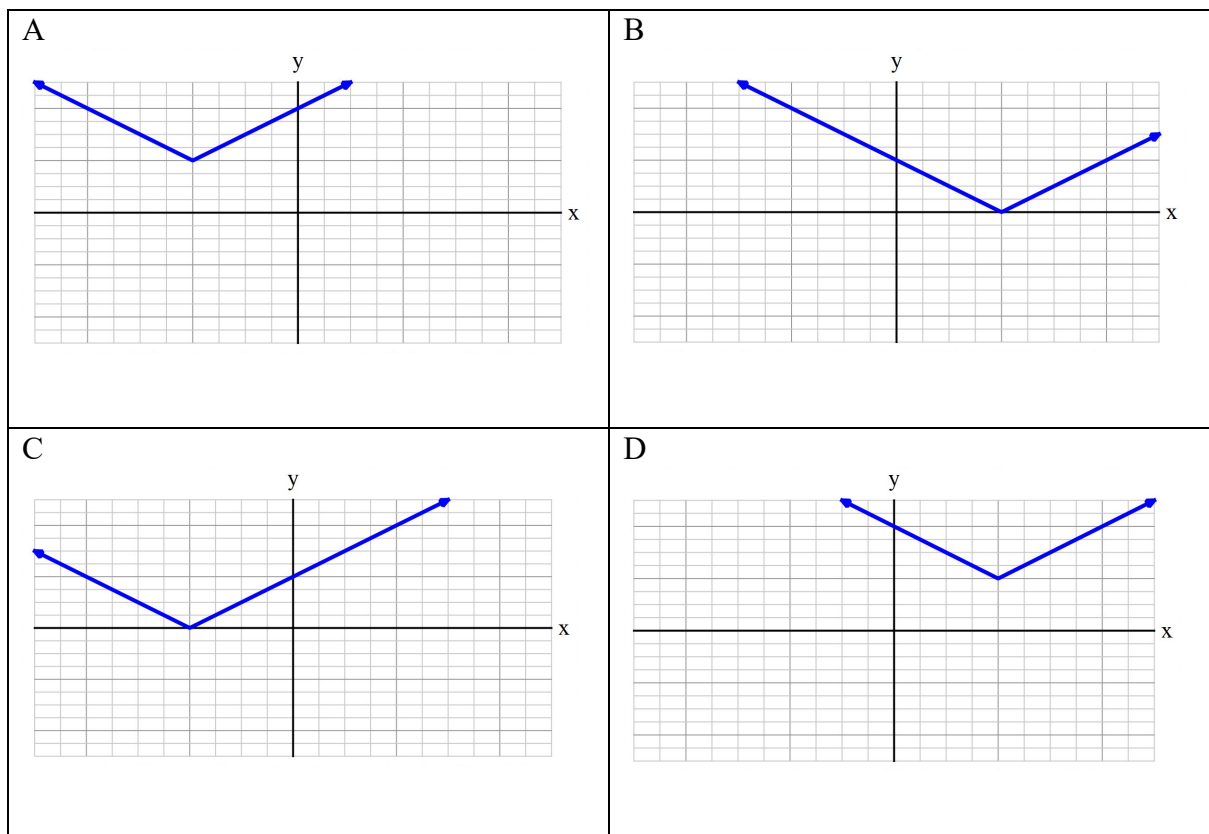
A. $\frac{dy}{dx} = \frac{5}{7 - 4t}$

B. $\frac{dy}{dx} = \frac{7 - 4t}{5}$

C. $\frac{dy}{dx} = 5 \times (7 - 4t)$

D. $\frac{dy}{dx} = \frac{1}{5(7 - 4t)}$

10. The graph of $y = |2 - x|$ is:



END OF SECTION A

SECTION B: QUESTIONS 11 TO 14. (15 MARKS EACH.)

PLEASE ANSWER EACH QUESTION ON SEPARATE BOOKLETS.

Question 11 Start on a new answer booklet

15 Marks

(a) Solve for x , given $\frac{1-2x}{1+x} \geq 1$ 3

(b) When a polynomial $P(x)$ is divided by $x^2 - 16$ the remainder is $3x - 1$.
What is the remainder when $P(x)$ is divided by $(x - 4)$. 2

(c) Find $\frac{d}{dx}(e^x \tan^{-1} x)$ 2

(d) The equation $I(t) = 4\sin 0.02t + 8\cos 0.02t$ measures the current, I amperes,
in a particular circuit after t seconds.

(i) Express the function in the form $I(t) = A\sin(at + b)$. for $b = \left[0, \frac{\pi}{2}\right]$ 2

(ii) When does the current first reach its maximum value to the nearest second? 2

(e) Find the equation of the curve $r = r(\theta)$, which satisfies the differential equation

$\frac{dr}{d\theta} = \frac{r^2}{\theta}$, with initial conditions $r(1) = 2$. 2

(f) Sketch $y = 2\sin^{-1}x + \frac{\pi}{2}$. Showing all intercepts. 2

End of Question 11

Question 12 Start on a new answer booklet**15 Marks**

(a) Factorise and solve $2x^3 + x^2 - 7x - 6 = 0$. 2

(b) Show that $\tan\left(\cos^{-1}\left(\frac{2}{3}\right) + \tan^{-1}\left(-\frac{3}{4}\right)\right) = \frac{4\sqrt{5} - 6}{3\sqrt{5} + 8}$. 3

(c) Prove by the principal of Mathematical Induction for all integers $n \geq 1$.

$$x + x^3 + x^5 + \dots + x^{2n-1} = \frac{x(x^{2n} - 1)}{(x - 1)(x + 1)}.$$
 3

(d) ΔABC has vertices $A(-4,7)$, $B(3,6)$ and $C(-5,0)$.

(i) Express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ in column vector form. 2

(ii) Find the magnitude of $\overrightarrow{AB}$ and $\overrightarrow{AC}$. 2

(iii) Find $\angle BAC$. 2

(iv) Hence calculate the area of ΔABC . 1

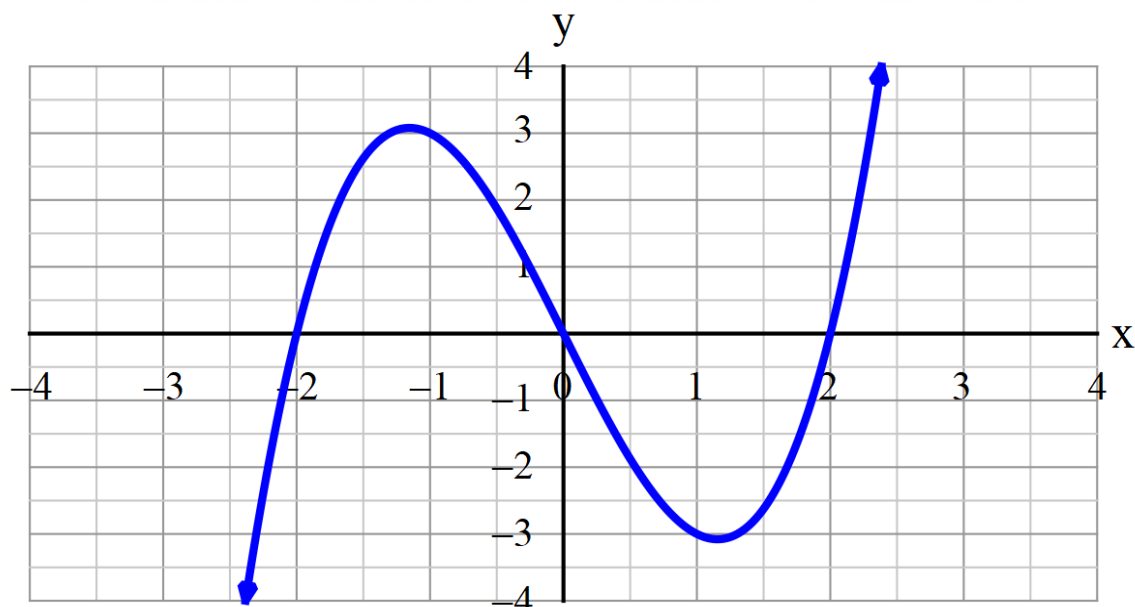
End of Question 12

Question 13 Start on a new answer booklet

15 Marks

(a) The diagram below shows the graph of a function $y = f(x)$.

The function has zeros at $x = -2$, $x = 0$ and $x = 2$ and a y-intercept of 0 as shown.



(i) Sketch the graph of $y = f(x)$ and $y = |f(x)|$ on the same axis.
Showing all essential features: intercepts, asymptotes.

1

(ii) Sketch the graph of $y = f(x)$ and $y = \frac{1}{f(x)}$ on the same axis.
Showing all essential features: intercepts, asymptotes.

3

Your graph in both cases should be at least half a page each.

(b) The graph of the function $y = \sin^{-1}(x - 3)$ is transformed by being dilated by a factor 3 horizontally and then translated to the right by 3 units.
Find the equation of the transformed path.

2

(c) (i) Find $\frac{d}{dx} \left[\frac{2x}{4 + x^2} + \tan^{-1} \frac{x}{2} \right]$,

write your answer in the form $\frac{A}{(4 + x^2)^2}$

2

(ii) Hence evaluate $\int_0^2 \frac{dx}{(4 + x^2)^2}$.

2

- (d) A fast food chain has estimated that if they spend x on advertising their new product. It will attract a proportion $y(x)$ of the potential customers for the product.

where $\frac{dy}{dx} = ay(1 - y)$. $a > 0$ is a given constant.

(i) Show that $\frac{1}{y} + \frac{1}{1 - y} = \frac{1}{y(1 - y)}$. 1

(ii) Using part (i) or otherwise,

show that $y(x) = \frac{1}{ke^{-ax} + 1}$ for some constant $k > 0$. 3

- (iii) The fast food chain knows that if they do not spend the money on advertising the product then the number of customers will be one-fifth of the potential customers. Hence find the value of the constant k in part (ii). 1

End of Question 13

Question 14 Start on a new answer booklet**15 Marks**

- (a) Find the exact value of $\tan\left(2\cos^{-1}\frac{12}{13}\right)$ 2

Showing all appropriate calculations.

- (b) (i) Use the substitution $t = \tan \frac{x}{2}$ to show that $\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$. 2

(ii) Hence or otherwise evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$.

Answer in the simplest exact form. 3

- (c) The points P , Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively.

λ and μ are non-zero numbers such that $\lambda \mathbf{p} + \mu \mathbf{q} - \mathbf{r} = \mathbf{0}$ and $\lambda + \mu = 1$

- (i) Show that points P , Q and R are collinear. 2

- (ii) It is known that $|\mathbf{p}| = 2$, the angle between $\mathbf{p}$ and $\mathbf{q}$ is acute and the area of ΔOPQ is $k \text{ units}^2$.

Show that $(\mathbf{p} \cdot \mathbf{q})^2 = 4(|\mathbf{q}|^2 - k^2)$. 3

Point S lies on the line segment PR and has a position vector $\mathbf{s}$.

- (iii) Given that $k = 6$, $|\mathbf{q}| = 10$ and $\angle POS = 90^\circ$. Find $\mathbf{s}$ in terms of $\mathbf{p}$ and $\mathbf{q}$. 3

End of Question 14

End of Task 4

2025

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Student's Name: _____ Teacher's Code: _____



SUGGESTED SOLUTIONS

Saint Ignatius' College, Riverview

Mathematics Assessment Task

2025

Year 12
Mathematics (Extension One)
Task 4
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Section B

Short Answer	
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Total	70 Marks
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Start your answer here.

$$1. y = (x-3)^3(x-2)(x+4)^4$$

Roots: $x=3$ +, multiplicity = 2

$$x=2 \text{ +}, \quad \text{"} = 1$$

$$x=-4 \text{ -}, \quad \text{"} = 4$$

$\therefore D.$

$$2. \text{ For scalar projection } \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{11}{\sqrt{13}}$$

$$|\vec{b}|: A=\sqrt{13}, B=\sqrt{13}, C=\sqrt{28}, D=\sqrt{25}$$

$\therefore$ Either A or B.

$$\text{Check A: } \vec{a} \cdot \vec{b} = (1)(2) + (4)(3) = 14 \times$$

$$B: \vec{a} \cdot \vec{b} = (1)(3) + (4)(2) = 11 \checkmark$$

$\therefore B.$

$$3. \sin 75^\circ = \sin (45^\circ + 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3}}{4} + \frac{1}{2\sqrt{2}}$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$\therefore \textcircled{A}$

$$\begin{aligned}
 4. \cos^{-1}\left(-\frac{1}{2}\right) &= \pi - \cos^{-1}\left(\frac{1}{2}\right) \\
 &= \pi - \frac{\pi}{3} \\
 &= \frac{2\pi}{3}
 \end{aligned}$$

$\therefore C$

$$5. \int \sin 3x \sin x dx \quad \text{use formula}$$

$\sin A \sin B$

$$= \frac{1}{2} [\cos(3x-x) - \cos(3x+x)]$$

$$= \frac{1}{2} \cos 2x - \cos 4x$$

$$\therefore = \frac{1}{2} \int \cos 2x - \cos 4x dx$$

$$= \frac{1}{2} \frac{\sin 2x}{2} - \frac{1}{2} \frac{\sin 4x}{4} + C$$

$$= \frac{\sin 2x}{4} - \frac{\sin 4x}{8} + C$$

$\therefore A$

$$6. x^2 + 2x + 3 = 0$$

$$\alpha + \beta = -2 \quad \alpha\beta = 3$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-2}{3}$$

$\therefore B$

Additional writing space on back page.

y

$$7. y = \frac{3 + e^{2x}}{5}$$

$$\therefore x = \frac{3 + e^{2y}}{5}$$

$$3 + e^{2y} = 5x$$

$$e^{2y} = 5x - 3$$

$$2y = \ln|5x - 3|$$

$$y = \frac{1}{2} \ln|5x - 3|$$

$\therefore B$

$$8. \vec{p} \cdot \vec{q} = |\vec{p}| |\vec{q}| \cos \theta = 3 \times 3 \times \cos 60^\circ$$

$$= \frac{9}{2}$$

$\therefore D$

$$9. x = 40t, y = 56t - 16t^2$$

$$dx = 40dt, dy = (56 - 32t)dt$$

$$\therefore \frac{dy}{dx} = \frac{(56 - 32t)dt}{40dt} = \frac{8(7 - 4t)}{5}$$

$\therefore B$

$$10. \text{Graph: } x=2, y=0$$

$\therefore B$

You may ask for an extra Writing Booklet if you need more space to answer this question.

Student's Name: Solutions

Teacher's Code: Ext 1: Trials 2025

Question Number	Marker use only
	Q 11



Saint Ignatius' College
RIVERVIEW

Mathematics Extension 1

Writing Booklet

(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	

Instructions

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Start your answer here.

✓ (a) $\frac{1-2x}{1+x} \geq 1$ $1+x \neq 0 \therefore x \neq -1$

$$\therefore \frac{(1-2x)(1+x)}{(1+x)^2} \geq 1$$

$$(1-2x)(1+x) \geq (1+x)^2$$

$$(1-2x)(1+x) - (1+x)^2 \geq 0$$

$$(1+x)[1-2x-1-x] \geq 0$$

$$(1+x)(-3x) \geq 0$$

$$3x(1+x) \leq 0$$



$$-1 < x \leq 0 \quad \text{or} \quad (-1, 0]$$

✓ (b) $(x^2-16) \cdot 8(x) + (3x-1)$

$$P(4) = 0 \cdot 8(x) + 3 \times 4 - 1$$

$$= 11$$

✓ (c) $\frac{d}{dx}(e^x \tan^{-1} x)$

$$u = e^x \quad v = \tan^{-1} x$$

$$u' = e^x \quad v' = \frac{1}{1+x^2}$$

$$\therefore e^x \tan^{-1} x + e^x \times \frac{1}{1+x^2}$$

$$\text{or } e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right)$$

Q 11(d)

(i) $I(t) = 4\sin(0.02t) + 8\cos(0.02t)$

$$A = \sqrt{4^2 + 8^2} = \sqrt{80} = 4\sqrt{5}$$

$$\tan \theta = \frac{8}{4} \quad \theta = \tan^{-1}\left(\frac{8}{4}\right) = \tan^{-1} 2$$

$$I(t) = 4\sqrt{5} \sin(0.02t + \tan^{-1} 2)$$

(ii) Max. Value at $\sin \theta = \frac{\pi}{2}$

$$\therefore 0.02t + \tan^{-1} 2 = \frac{\pi}{2}$$

$$0.02t = \frac{\pi}{2} - \tan^{-1} 2$$

$$t = \frac{\frac{\pi}{2} - \tan^{-1} 2}{0.02} = 23 \text{ seconds.}$$

(18 max)

Start your answer here.

$$11 \text{ (e) } \frac{dr}{d\theta} = \frac{r^2}{\theta}$$

$$\therefore \int \frac{1}{r^2} dr = \int \frac{d\theta}{\theta}$$

$$-\frac{1}{r} = \ln \theta + C$$

$$\theta = 1, r = 2$$

$$\therefore -\frac{1}{2} = \ln 1 + C$$

$$C = -\frac{1}{2}$$

$$\therefore -\frac{1}{r} = \ln |\theta| - \frac{1}{2}$$

$$\therefore \frac{1}{r} = \frac{1}{2} - \ln |\theta|$$

$$\frac{1}{r} = \frac{1 - 2 \ln |\theta|}{2}$$

$$r = \frac{2}{1 - 2 \ln |\theta|}$$

OR.

$$\frac{1}{r^2} dr = \frac{1}{\theta} d\theta$$

$$\therefore \int_2^r \frac{1}{r^2} dr = \int_1^\theta \frac{1}{\theta} d\theta$$

$$\left[-\frac{1}{r} \right]_2^r = \left[\ln |\theta| \right]_1^\theta$$

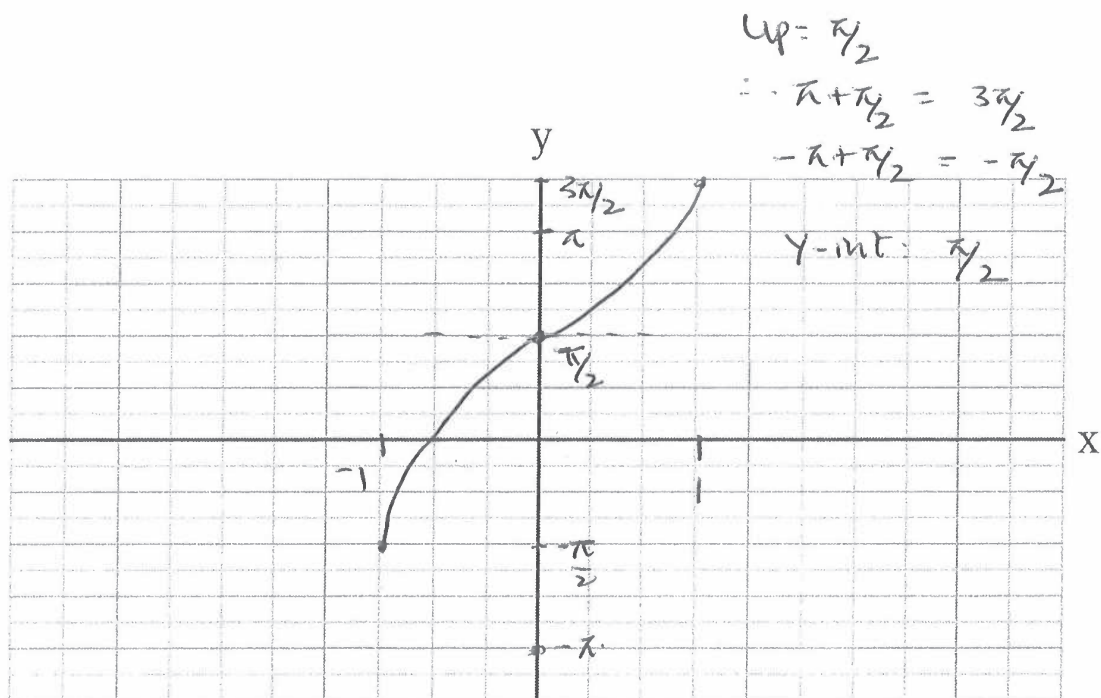
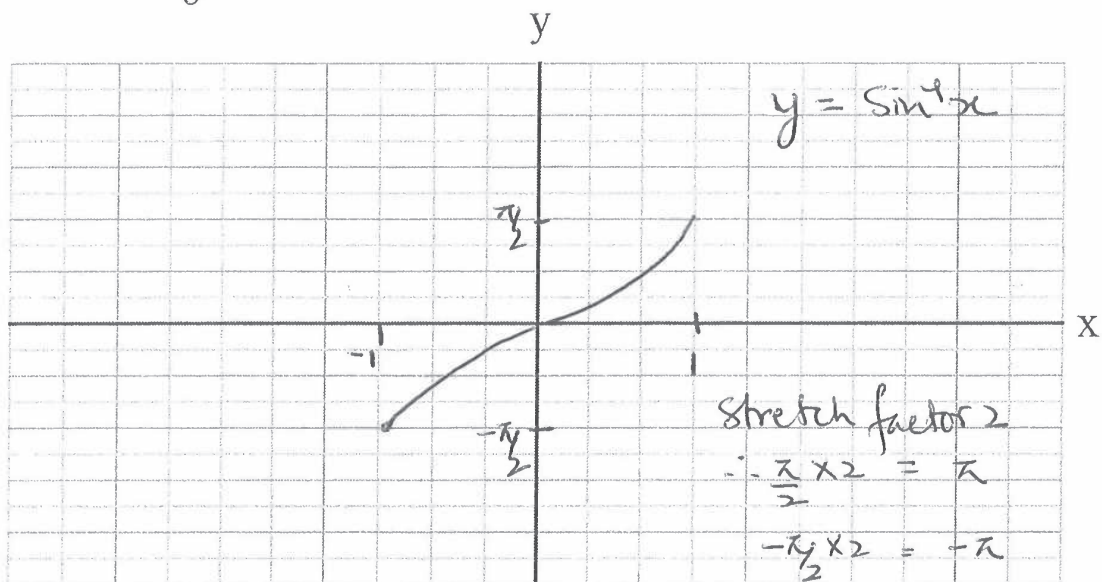
$$-\frac{1}{r} + \frac{1}{2} = \ln |\theta| - \ln 1$$

$$\frac{1}{r} - \frac{1}{2} = \ln |\theta|$$

$$\frac{1}{r} = \frac{1 - 2 \ln |\theta|}{2}$$

$$\therefore r = \frac{2}{1 - 2 \ln |\theta|}$$

(f) $y = 2\sin^2 x + \frac{\pi}{2}$



Student's Name: 812 Teacher's Code: Solutions

Question Number	Marker use only
	812



Saint Ignatius' College
RIVERVIEW

Mathematics Extension 1

Writing Booklet

(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	

Instructions

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Start your answer here.

Question 12

12(a) $2x^3 + x^2 - 7x - 6 = 0$

$P(-1) = 0 \quad \therefore (x+1) \text{ is a factor}$

$$2x^3 + 2x^2 - x^2 - x - 6x - 6$$

$$2x^2(x+1) - x(x+1) - 6(x+1)$$

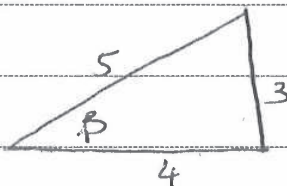
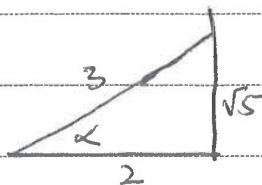
$$(x+1)(2x^2 - x - 6)$$

$$(x+1)(2x+3)(x-2) = 0$$

$$\therefore x = -1, x = -\frac{3}{2}, x = 2$$

12 (b) $\tan\left[\cos^{-1}\left(\frac{2}{3}\right) + \tan^{-1}\left(-\frac{3}{4}\right)\right]$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$



$$\tan \alpha = \frac{\sqrt{5}}{2}$$

$$\tan \beta = -\frac{3}{4} \quad (\text{II, IV Q})$$

$$\therefore \tan(\alpha + \beta) = \frac{\frac{\sqrt{5}}{2} + \left(-\frac{3}{4}\right)}{1 - \frac{\sqrt{5}}{2} \times \left(-\frac{3}{4}\right)}$$

$$= \frac{2\sqrt{5} - 3}{4}$$

$$\frac{8 + 3\sqrt{5}}{8}$$

$$= \frac{2\sqrt{5} - 3}{4} \times \frac{8}{8 + 3\sqrt{5}}$$

$$= \frac{2(2\sqrt{5} - 3)}{8 + 3\sqrt{5}}$$

$$\text{OR} = \frac{4\sqrt{5} - 6}{3\sqrt{5} + 8}$$

12 (c) Induction:

$$x + x^3 + x^5 + \dots + x^{2n+1} = \frac{x(x^{2n+1} - 1)}{(x-1)(x+1)}$$

Prove for $n=1$

$$\text{LHS: } x^{2(1)+1} = x^3$$

$$\text{RHS: } \frac{x(x^{2(1)+1} - 1)}{(x-1)(x+1)} = \frac{x(x^3 - 1)}{(x^2 - 1)} = x$$

$$\therefore \text{LHS} = \text{RHS}$$

Assume true for $n=k$.

$$\therefore x + x^3 + x^5 + \dots + x^{2k+1} = \frac{x(x^{2k+1} - 1)}{(x-1)(x+1)}$$

Prove for $n=k+1$.

$$\therefore x + x^3 + x^5 + \dots + x^{2k+1} + x^{2k+3} = \frac{x(x^{2k+1} - 1)}{(x-1)(x+1)} + x^{2k+3} = \frac{x(x^{2k+3} - 1)}{(x^2 - 1)}$$

By Assumption LHS

$$\frac{x(x^{2k} - 1)}{(x^2 - 1)} + x^{2k+3}$$

$$\frac{x(x^{2k} - 1) + x^{2k+3}(x^2 - 1)}{(x^2 - 1)}$$

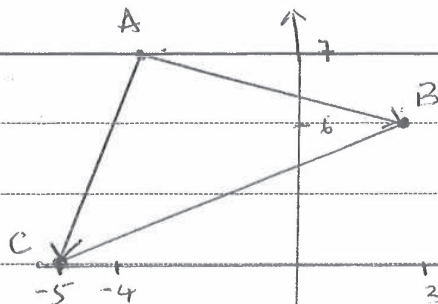
$$\frac{x \cdot x^{2k} - x + x^{2k+3} \cdot x^2 - x^{2k+3}}{(x^2 - 1)}$$

$$= \frac{x(x^{2k+3} - 1)}{(x^2 - 1)} = \text{RHS}$$

$\therefore$ True by mathematical induction.

Additional writing space on back page.

d.



$$(i) \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} -4 \\ 7 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$

$$= \begin{pmatrix} -5 \\ 0 \end{pmatrix} - \begin{pmatrix} -4 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \end{pmatrix}$$

$$(ii) |\overrightarrow{AB}| = \sqrt{7^2 + (-1)^2} = \sqrt{50} = 5\sqrt{2}$$

$$|\overrightarrow{AC}| = \sqrt{(-1)^2 + (-7)^2} = \sqrt{50} = 5\sqrt{2}$$

$$(iii) \overrightarrow{AB} \cdot \overrightarrow{AC} = |\overrightarrow{AB}| |\overrightarrow{AC}| \cos \theta$$

$$(7)(-1) + (-1)(-7) = 5\sqrt{2} \times 5\sqrt{2} \cos \theta$$

$$-7 + 7 = 50 \cos \theta$$

$$0 = 50 \cos \theta$$

$$\therefore \cos \theta = 0 \quad \theta = \frac{\pi}{2}$$

$$\angle BAC = \frac{\pi}{2}$$

Additional writing space on back page.

(iv) Area $\triangle ABC$ (Since rt. $\triangle$)

$$\begin{aligned} & \frac{1}{2} b \times h \\ &= \frac{1}{2} \times 5\sqrt{2} \times 5\sqrt{2} \\ &= 25u^2 \end{aligned}$$

$$\begin{aligned} \text{OR: } A &= \frac{1}{2} ab \sin C \quad (\text{Formula}) \\ &= \frac{1}{2} \times 5\sqrt{2} \times 5\sqrt{2} \sin \frac{\pi}{2} \\ &= 25u^2. \end{aligned}$$

Student's Name: Soluhina Teacher's Code: Q13

Question Number	Marker use only
	Q 13



Saint Ignatius' College
RIVERVIEW

Mathematics Extension 1

Writing Booklet

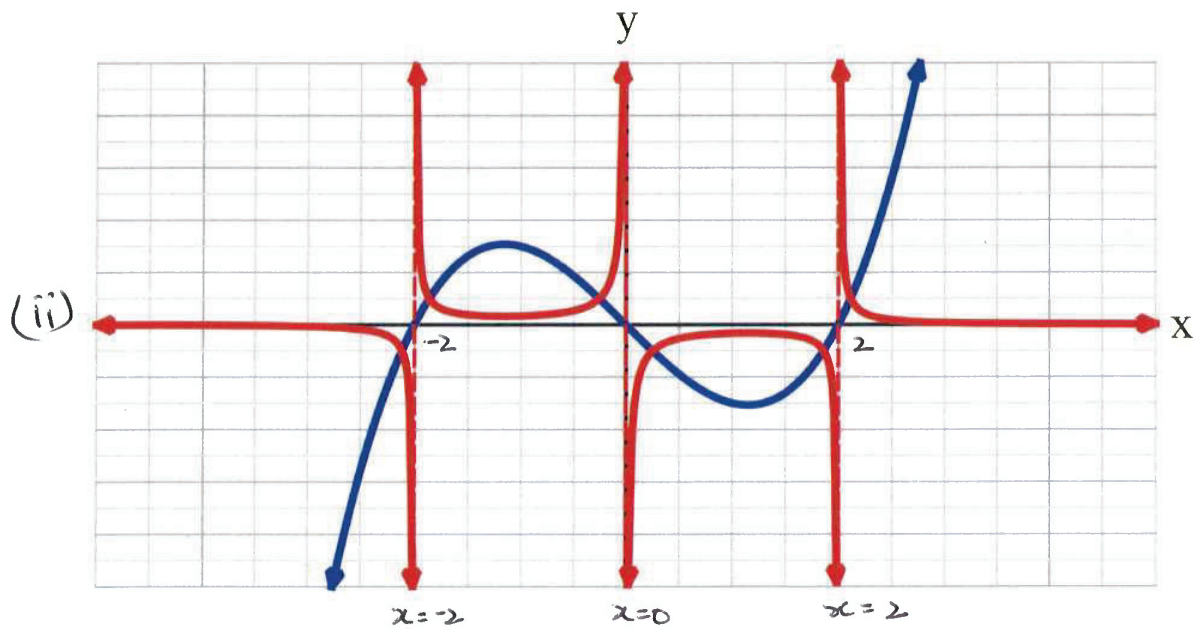
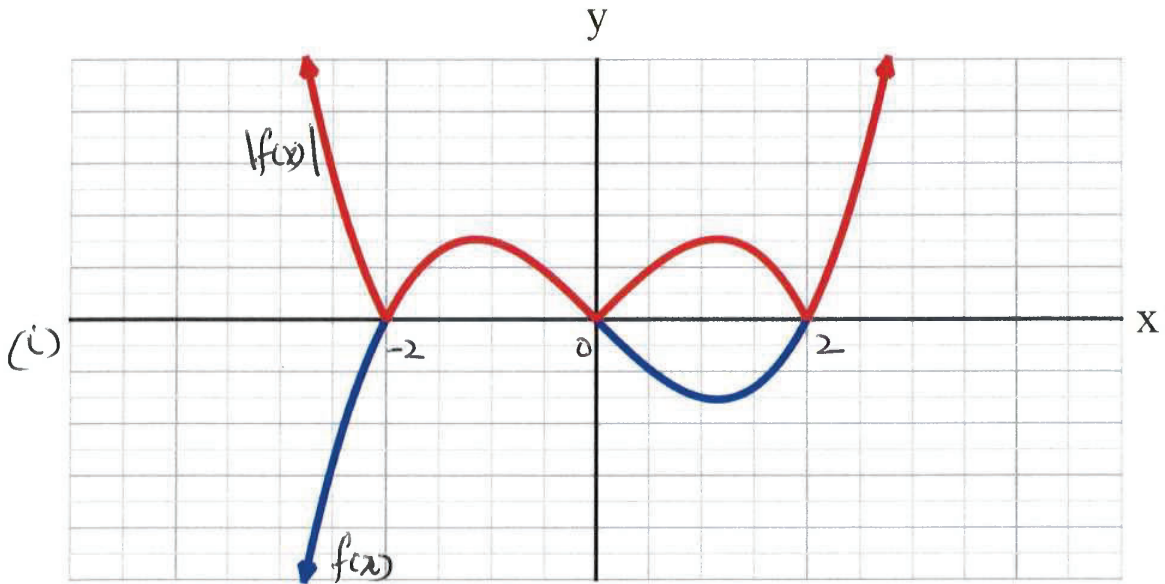
(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	

Instructions

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Question 13 (a)



Asymptotes at $x = \pm 2$, $x = 0$ and $y = 0$

Start your answer here.

13 (b)

$$y = \sin^{-1}(x-3)$$

Dilated by a factor 3 from the y-axis

$$x \rightarrow \frac{x}{3}$$

$$\therefore y = \sin^{-1}\left(\frac{x}{3} - 3\right) = \sin^{-1}\left(\frac{x-9}{3}\right)$$

Translate to the right by 3

$$x \rightarrow x-3$$

$$\therefore y = \sin^{-1}\left[\frac{(x-3)-9}{3}\right]$$

$$= \sin^{-1}\left(\frac{x-3-9}{3}\right)$$

$$y = \sin^{-1}\left(\frac{x-12}{3}\right) \quad \text{Final Ans.}$$

13 (6) (i)

$$\frac{d}{dx} \left[\frac{2x}{4+x^2} \right]$$

$$u = 2x \quad v = 4+x^2$$

$$u' = 2 \quad v' = 2x$$

$$= \frac{2(4+x^2) - 2x(2x)}{(4+x^2)^2}$$

$$= \frac{8+2x^2-4x^2}{(4+x^2)^2}$$

$$= \frac{8-2x^2}{(4+x^2)^2}$$

$$\frac{d}{dx} \left[\frac{\tan^{-1} x}{2} \right] = \frac{12}{1+(\frac{x}{2})^2} \times \frac{1}{2}$$

$$= \frac{1}{4+x^2} \times \frac{1}{2}$$

$$= \frac{2}{4+x^2}$$

$$\therefore \frac{d}{dx} \left[\frac{2x}{4+x^2} + \frac{\tan^{-1} x}{2} \right] = \frac{8-2x^2}{(4+x^2)^2} + \frac{2}{4+x^2}$$

$$= \frac{8-2x^2+2(4+x^2)}{(4+x^2)^2}$$

$$= \frac{8-2x^2+8+2x^2}{(4+x^2)^2}$$

$$= \frac{16}{(4+x^2)^2} = \frac{A}{(4+x^2)^2} \quad \text{Form.}$$

Additional writing space on back page.

(ii)

$$\int_0^2 \frac{dx}{(4+x^2)^2} = \frac{1}{16} \int \frac{16}{(4+x^2)^2} dx$$

$$= \frac{1}{16} \left[\frac{2x}{4+x^2} + \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= \frac{1}{16} \left[\frac{4}{4+4} + \tan^{-1} 1 - 0 - \tan^{-1} 0 \right]$$

$$= \frac{1}{16} \left[\frac{4}{8} + \frac{\pi}{4} \right]$$

$$= \frac{1}{32} + \frac{\pi}{64}$$

13(d)

$$(i) \frac{1}{y} + \frac{1}{y+1} =$$

$$\frac{1}{y} + \frac{1}{1-y} = \frac{1-y+y}{y(1-y)} = \frac{1}{y(1-y)}$$

$$(ii) \frac{dy}{dx} = ay(1-y)$$

$$\therefore \frac{dy}{y(1-y)} = a dx$$

$$\therefore \int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy = \int a dx \text{ from (ii)}$$

$$\ln|y| - \ln|1-y| = ax + c$$

$$\begin{aligned} -ax - c &= \ln|1-y| - \ln|y| \\ &= \ln \left| \frac{1-y}{y} \right| \end{aligned}$$

$$e^{-ax-c} = \frac{1-y}{y}$$

$$e^{-ax-c} = \frac{1}{y} - 1$$

$$e^{-ax-c} + 1 = \frac{1}{y}$$

$$y = \frac{1}{e^{-ax-c} + 1}$$

$$\frac{e^{-ax} \cdot e^{-c}}{K e^{-ax}}$$

$$y = \frac{1}{K e^{-ax} + 1}$$

As reqd.

Additional writing space on back page.

(iv)

$$y(x) = \frac{1}{Ke^{-ax} + 1}$$

Given $x=0$, $y(x) = \frac{1}{5}$

$$\therefore \frac{1}{5} = \frac{1}{K+1}$$

$$K+1 = 5$$

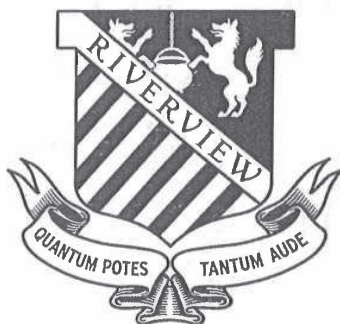
$$K = 4$$

You may ask for an extra Writing Booklet if you need more space to answer this question.

Student's Name: Question 14

Teacher's Code: Solutions

Question Number	Marker use only
	Q14



Saint Ignatius' College
RIVERVIEW

Mathematics Extension 1

Writing Booklet

(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	

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14 (10)

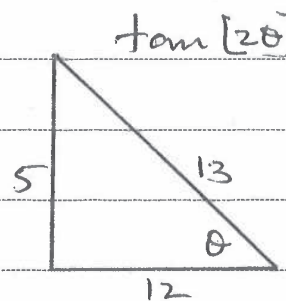
$$\tan\left[2\cos^{-1}\frac{12}{13}\right]$$

$$= \frac{2\tan\left[\cos^{-1}\frac{12}{13}\right]}{1 - \tan^2\left[\cos^{-1}\frac{12}{13}\right]}$$

$$= \frac{2\left(\frac{5}{12}\right)}{1 - \left(\frac{5}{12}\right)^2}$$

$$= \frac{120}{144 - 25}$$

$$= \frac{120}{119} \checkmark$$



formula.

$$= \frac{2\tan\theta}{1 - \tan^2\theta}$$

Start your answer here.

$$Q14 (b) \quad t = \tan \frac{x}{2}$$

$$(i) \therefore \text{LHS: } \operatorname{cosec} x + \cot x$$

$$\frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$

$$= \frac{1+t^2+1-t^2}{2t}$$

$$= \frac{2t}{2t} = \frac{1}{t}$$

$$= \cot \frac{x}{2} = \text{RHS.}$$

$$(ii) \int_{\pi/3}^{\pi/2} (\operatorname{cosec} x + \cot x) dx$$

$$= \int_{\pi/3}^{\pi/2} \cot \frac{x}{2} dx = \frac{1}{2} \int_{\pi/3}^{\pi/2} \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} dx$$

$$= 2 \left[\ln \left(\sin \frac{x}{2} \right) \right]_{\pi/3}^{\pi/2}$$

$$= 2 \left[\ln \left(\sin \frac{\pi}{4} \right) - \ln \sin \left(\frac{\pi}{6} \right) \right]$$

$$= 2 \left[\ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right] = 2 \ln \left(\frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} \right)$$

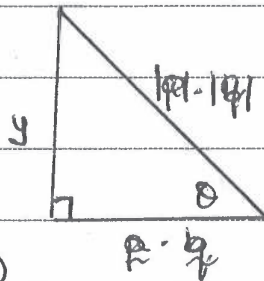
$$= 2 \ln 2^{1/2} = \ln 2.$$

(c)

$$\begin{aligned}\text{(i)} \quad \vec{PQ} &= \vec{q} - \vec{p}, \quad \vec{PR} = \vec{r} - \vec{p} \\ &= \lambda \vec{p} + \mu \vec{q} - \vec{p} \\ &= (\lambda - 1)\vec{p} + \mu \vec{q} \\ &= -\mu \vec{p} + \mu \vec{q} \\ &= \mu(\vec{q} - \vec{p}) \\ \therefore \vec{PR} &= \mu \vec{PQ}\end{aligned}$$

$\therefore P, Q, R$ are collinear.

(ii)



$$y^2 = (|p||q|)^2 - (P \cdot q)^2$$

$$K = \frac{1}{2} |p||q| \sin \theta \quad (\text{Given})$$

$$2 \cdot K = \frac{1}{2} |p||q| \times \frac{\sqrt{(|p||q|)^2 - (P \cdot q)^2}}{|p||q|}$$

$$\text{sq.} \quad 4K^2 = (|p||q|)^2 - (P \cdot q)^2$$

$$\begin{aligned}(P \cdot q)^2 &= (|p||q|)^2 - 4K^2 \\ &= 4|q|^2 - 4K^2 \\ &= 4(|q|^2 - K^2)\end{aligned}$$

$$|p| = 2$$

Start your answer here.

(c)

$$\begin{aligned} \text{(iii)} \quad (\underline{p} - \underline{q})^2 &= 4 [\underline{q}^2 - \underline{p}^2] \\ &= 4 [10^2 - 6^2] \\ &= 256 \end{aligned}$$

$$\therefore \underline{p} \cdot \underline{q} = 16$$

Since D lies on $\overline{PQ}$, then

$$\underline{OD} = \underline{OA} + \lambda \underline{AB} \quad \lambda \in \mathbb{R}$$

$$\underline{s} = \underline{p} + \lambda(\underline{q} - \underline{p})$$

$$\text{and } \angle POQ = 90^\circ \quad \therefore \underline{p} \cdot \underline{s} = 0$$

$$\underline{p} [\underline{p} + \lambda(\underline{q} - \underline{p})] = 0$$

$$|\underline{p}|^2 + \lambda \underline{p} \cdot \underline{q} - \lambda |\underline{p}|^2 = 0$$

$$|\underline{p}|^2(1 - \lambda) + \lambda \underline{p} \cdot \underline{q} = 0$$

$$4(1 - \lambda) + 16\lambda = 0$$

$$\lambda = \frac{1}{3}$$

$$\underline{s} = \underline{p} - \frac{1}{3}(\underline{q} - \underline{p})$$

$$\underline{s} = \frac{4}{3}\underline{p} - \frac{1}{3}\underline{q}$$

Start your answer here.

Alternate Soln

~~14(c)~~

14(c)

$$(i) \quad \overrightarrow{PQ} = \underline{q} - \underline{p}$$

$$\overrightarrow{PR} = \underline{r} - \underline{p}$$

$$\begin{aligned} \overrightarrow{PR} &= \lambda \underline{p} + \mu \underline{q} - \underline{p} \\ &= (\lambda - 1) \underline{p} + \mu \underline{q} \\ &= -\mu \underline{p} + \mu \underline{q} \end{aligned}$$

$$\therefore \overrightarrow{PR} = \mu (\underline{q} - \underline{p})$$

$\therefore P, Q, R$ collinear.

Given:

$$\lambda \underline{p} + \mu \underline{q} - \underline{r} = \underline{0}$$

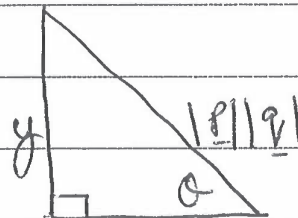
$$\lambda + \mu = 1$$

$$\therefore \mu = 1 - \lambda$$

$$\boxed{-\mu = \lambda - 1}$$

(ii)

$$\therefore \underline{p} \cdot \underline{q} = |\underline{p}| |\underline{q}| \cos \theta$$



$$\text{or } y^2 = (|\underline{p}| |\underline{q}|)^2 - (\underline{p} \cdot \underline{q})^2 \quad \text{By thm. } \underline{p} \cdot \underline{q}$$

$$K = \frac{1}{2} |\underline{p}| |\underline{q}| \sin \theta \quad \text{Area of } \Delta$$

$$K = \frac{1}{2} \cancel{|\underline{p}| |\underline{q}|} \times \frac{\sqrt{(|\underline{p}| |\underline{q}|)^2 - (\underline{p} \cdot \underline{q})^2}}{\cancel{|\underline{p}| |\underline{q}|}}$$

$$4K^2 = (|\underline{p}| |\underline{q}|)^2 - (\underline{p} \cdot \underline{q})^2$$

$$\begin{aligned} (\underline{p} \cdot \underline{q})^2 &= (|\underline{p}| |\underline{q}|)^2 - 4K^2 \\ &= 4|\underline{q}|^2 - 4K^2 \\ &= 4(|\underline{q}|^2 - K^2) \end{aligned}$$

$$\underline{p} = 2$$

$$(iii) (\underline{p} \cdot \underline{q})^2 = 4(\underline{p}^2 - k^2)$$

$$= 4(10^2 - 6^2)$$

$$= 256$$

$$\therefore \underline{p} \cdot \underline{q} = 16$$

Since $\underline{S}$ lies on $\overline{PR}$,

$$\text{then } \underline{OS} = \underline{OP} + \lambda \underline{PQ}$$

$$(*) \quad \underline{S} = \underline{P} + \lambda(\underline{q} - \underline{P}) \quad \lambda \in \mathbb{R}$$

If $\angle POS = 90^\circ$ then $\underline{P} \cdot \underline{S} = 0$

$$\therefore \underline{P} \cdot (\underline{P} + \lambda(\underline{q} - \underline{P})) = 0$$

$$\underline{P}^2 + \lambda \underline{P} \cdot \underline{q} - \lambda \underline{P}^2 = 0$$

$$\underline{P}^2 = \underline{P}^2$$

$$\underline{P}^2(1 - \lambda) + \lambda \underline{P} \cdot \underline{q} = 0$$

$$\underline{P} \cdot \underline{q} = 16$$

$$4(1 - \lambda) + 16\lambda = 0$$

$$\underline{P}^2 = 4$$

$$4 - 4\lambda + 16\lambda = 0 \quad \text{cancel } \lambda$$

$$12\lambda + 4 = 0 \quad \lambda = -\frac{1}{3}$$

$$\therefore \underline{S} = \underline{P} + \lambda(\underline{q} - \underline{P})$$

$$= \underline{P} - \frac{1}{3}(\underline{q} - \underline{P})$$

$$\text{or } \frac{4}{3}\underline{P} - \frac{1}{3}\underline{q}$$